EFY TESTED
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A Small String Of DECORATIVE FLASHING LIGHTS Using Arduino

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During festivals like Diwali, Christmas, and New Year strings of flashing lights are sometimes used to decorate homes and other premises. Arduino based flashing lights can be made with the help of just a few components for such occasions. The author's circuit wired on a breadboard for this purpose is shown in Fig. 1.

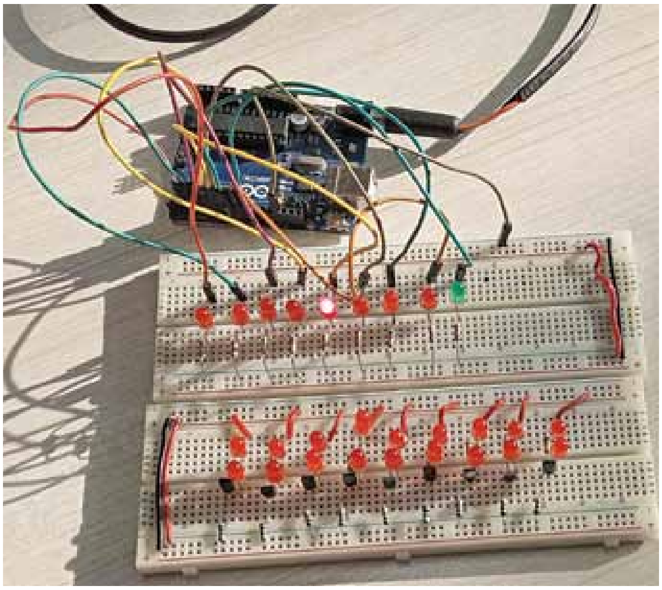

Fig. 1: Author's prototype on breadboard

The circuit diagram of Arduino based flashing light is shown in Fig. 2. The circuit comprises an Arduino Uno board (BOARD1), nine LEDs (LED1 through LED9) to demonstrate the concept, and a few discrete components only.

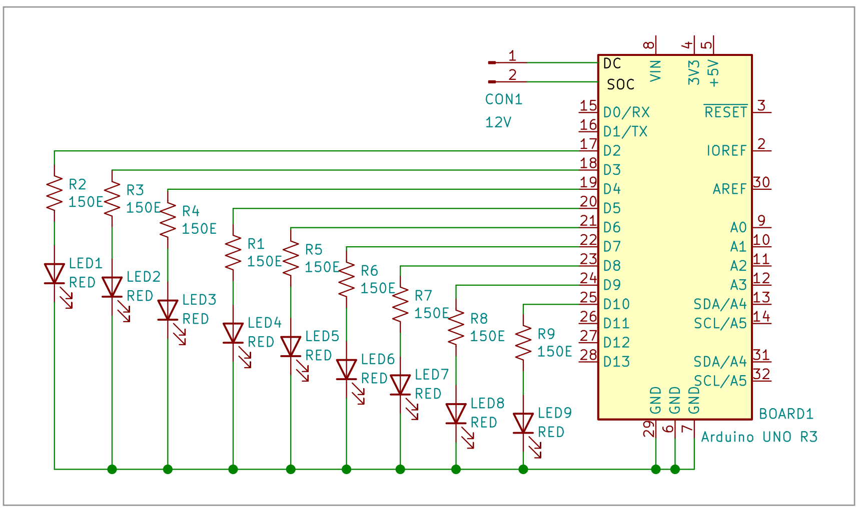

Fig. 2: Circuit diagram for the flashing light

The nine output pins of the Uno board are connected to the nine LEDs, one each, which switch on and switch off at predefined interval. As the high state of an output pin shifts from one pin to another, only one output pin is high at a time and switches on the LED connected to it. Thus, it looks like a running light, which can be arranged in a circle or any other desired shape.

To make a longer string of flashing LEDs, you may use the circuit shown in Fig. 3. This circuit flashes 18 LEDs, for which it makes use of nine NPN transistors in addition to the other components.

In the 18-LED flasher, nine outputs of Arduino Uno are connected to the bases of transistors T1 through T9 via resistors R1 through R9. The outputs go high and low after a predefined time. As before, the high state output shifts from one pin to the other but only one output pin is high at a time. Thus, the output again produces a running light effect.

Resistors R1 through R9, in both the circuits, are used to limit the current passing through each of the LEDs connected to them. To make a longer string, you can experiment with more LEDs in series and reduce the values of resistors R1 through R9 accordingly.

Software

The simple software for this project is written in Arduino IDE version 1.8.5. You can however use any version as it is a very simple program.